Ariyah Walker

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EDUCATION

University of Virginia **Charlottesville, VA**
Bachelor of Arts, Applied Statistics and Biology May 2026

- Honors: Mu Sigma Rho | Tri-Alpha Honor Society | Dean's List Spring 2025, 2026

Relevant Coursework: Mathematical Statistics · Statistical Machine Learning · Data Analysis with Python · Regression Analysis · Applied Linear Models · Data Visualization & Management · Data to Knowledge · Linear Algebra

PROFESSIONAL EXPERIENCE

BioKind Analytics **Charlottesville, VA**
Data Analyst, Healthcare Nonprofit Consulting September 2025 – April 2026

- Consolidated, cleaned, and merged patient, survey, and financial data from multiple Excel sources using R, creating an analysis-ready dataset of 10k+ records for healthcare outcome and cost analyses for nonprofit healthcare organizations.
- Developed and validated predictive logistic regression classification models in R using train/test splits and cross-validation, achieving 80–85% accuracy and identifying factors associated with successful health treatment outcomes.
- Applied general linear models, AIC-based variable selection, and regression diagnostics to identify treatment cost drivers and support data-driven recommendations to reduce expenses while maintaining quality of care.
- Analyzed healthcare cost drivers using statistical modeling and regression diagnostics, supporting data-driven recommendations to improve resource allocation while maintaining quality of care.
- Created data visualizations, reports, and presentations in R (ggplot2) and Tableau, translating complex datasets into actionable recommendations for stakeholders and decision-makers.

Dominion Energy **Richmond, VA**
Intern, Environmental Services Department June 2021 - August 2021

- Maintained environmental monitoring datasets across five project records, standardizing terminology and unified information from multiple departments to improve reporting efficiency and data consistency.
- Utilized Excel formulas, lookup functions, and Power BI visualizations to organize, analyze, and report environmental data, supporting project tracking and operational decision-making.
- Tracked activities across three collaborating teams and provided supervisors with progress updates to support project coordination and timely task completion.
- Organized environmental records and compliance documentation to support cross-functional communication and maintain accurate project information.

RESEARCH EXPERIENCE

Workforce Demand & Education Analytics

- Collected, scraped, and cleaned workforce, education, and economic datasets from multiple public sources using SQL, Python, and R, creating a unified dataset to evaluate labor market demand and degree production trends across Virginia.
- Designed an interactive R Shiny dashboard and data visualizations using ggplot2 and Plotly, enabling users to explore workforce shortages, educational attainment, salary outcomes, and employment trends.
- Conducted statistical analysis, exploratory data analysis, and data visualization in Python and R to identify high-demand occupations and workforce gaps, generating insights to support workforce planning and policy discussions.
- Synthesized complex workforce data into reports and presentations, translating technical findings into actionable recommendations for technical and non-technical audiences.

Air Quality and Emergency Department Visits Analysis

- Integrated and cleaned environmental, public health, and socioeconomic datasets from multiple public sources to evaluate factors associated with respiratory-related emergency department visits across Virginia.
- Applied exploratory data analysis, data transformations, and regression diagnostics to assess model assumptions and improve analytical accuracy.
- Developed and compared Poisson and Negative Binomial regression models in R, identifying seasonal patterns and socioeconomic factors as key predictors of respiratory-related ER visit volume.
- Created visualizations and presented findings demonstrating how air quality, seasonal trends, and community characteristics interact to influence healthcare utilization

SKILLS

Technical Tools: Python, R, SQL, Excel, Google Sheets, Tableau, Power BI, Microsoft Office, Git, GitHub, Unix/Linux, SLURM
Statistical Methods: Logistic Regression, Classification, Exploratory Data Analysis, Regression Diagnostics, Predictive Modeling
Professional Skills: Research, Technical Writing, Stakeholder Communication, Cross-Functional Collaboration, Project Coordination, Analytical Problem-Solving, Attention to Detail